



SRI VASAVI ENGINEERING COLLEGE (AUTONOMOUS)

(Sponsored by Sri Vasavi Educational Society)

(Approved by AICTE, New Delhi & Permanently affiliated to JNTUK, Kakinada)

(Accredited by NAAC with 'A' Grade, Recognized by UGC under section 2(f) & 12(B))

Pedatadepalli, **TADEPALLIGUDEM – 534 101**, W.G.Dist. (A.P)

Department of Computer Science & Engineering (Accredited by NBA)

and

Department of Computer Science & Technology

Minutes of the Seventh Board of Studies held on 18/07/2024 at 11.00 AM through online mode

Date: 25.07.2022

The 7th Meeting of Board of Studies in Department of Computer Science and Engineering is held at 11:00 AM on 18-07-2024 through online mode using,

<https://us02web.zoom.us/j/82375941809>

The following members attended the meeting:

S.No.	Name of the Member	Designation	Role
1.	Dr. D Jaya Kumari	Professor, HoD-CSE, SVEC	Chairperson
2.	Dr.Krishna Mohan Ankala	Professor, UCEK, Kakinada	University Nominee
3.	Dr. R.B.V. Subramanyam	Professor, Department of CSE, NIT Warangal	Academic Expert
4.	Dr. S Pallam Setty	Professor, Department of CSE, Andhra University, Vishakapatnam	Academic Expert
5.	Sri. Srinivasa Raju Vuppalapati	Senior Consultant, MSR IT Services LLP, Hitech City, Hyderabad.	Industry Expert
6.	Mr.EEdala Rambabu	Amadues, Bangalore	Alumni
7.	Sri Ch. Apparao	Technical Director	Invited Member
8.	Dr. Ch Rambabu	I/C Principal, Professor of EEE	Invited Member
9.	Dr. G Loshma	Professor	HOD-CAI &AIM
10.	Dr. V S Naresh	Professor	Member
11.	Dr. Daveed Raju	Professor	Member
12.	Dr.K. Shirin Bhanu	Associate Professor	Member
13.	A Leelavathi	Sr. Assistant Professor	Member
14.	Mrs. B.Sri Ramya	Sr. Assistant Professor	Member
15.	N.V.Murali Krishna Raja	Sr. Assistant Professor	Member
16.	N. Hiranmayee	Sr. Assistant Professor	Member
17.	A Rajesh	Sr. Assistant Professor	Member
18.	K Lakshmi Narayana	Assistant Professor	Member
19.	Mr. M.V.V Krishna	Assistant Professor	Member
20.	P Uma Sankar	Assistant Professor	Member
21.	Mr. M.S.KumarReddy	Assistant Professor	Member
22.	P Rajesh	Assistant Professor	Member
23.	M Santhi	Assistant Professor	Member
24.	G Prashanthi	Assistant Professor	Member
25.	K Praveen Kumar	Assistant Professor	Member
26.	G Nagavallika	Assistant Professor	Member
27.	N V Ratna Kishore Gade	Assistant Professor	Member
28.	M N V Surekha	Assistant Professor	Member
29.	Mr. P. Ramamohan Rao	Assistant Professor	Member
30.	Mr. T. Anil Kumar Reddy	Assistant Professor	Member
31.	Mr. L. Atri Datta Ravi Tez	Assistant Professor	Member
32.	Mr. Md. Sadik	Assistant Professor	Member
33.	Ms. R. Nava Lavanya	Assistant Professor	Member
34.	Dr. G. Tej Varma	Assistant Professor	Member
35.	A Niranjana Rao	Assistant Professor	Member
36.	Mr. K. Phanindra Brahmaji	Lecturer	Member
37.	Mr. S. P. Ramesh Varma	Lecturer	Member

The following are the Minutes of the Meeting

Item No.1: Welcome note by the Chairperson BOS.

Chairperson BOS extended a formal welcome and introduced the members and had given the Brief Progress Report of the Department.

Item No.2: Approved Course Structure of I and II Semesters of B.Tech(CSE) and B.Tech(CST) Programme under V23 Regulation.

The Course Structure is given in **Annexure-I**.

Item No.3: Approval of Proposed Course Structure for III and IV Semesters of B.Tech(CSE) and B.Tech(CST) Programme under V23 Regulation

The Approved Course Structure is given in **Annexure-II**.

Item No. 4: Approval of Syllabi for Proposed Courses offered in III and IV Semesters of B.Tech(CSE) and B.Tech(CST) Programme under V23 Regulation

Approved the Syllabi for the courses offered in III and IV Semesters of B.Tech(CSE) and B.Tech(CST) Programme under V23 Regulation and suggested the following changes:

SEM	Name of the Course	Suggestions	Modifications
III	Digital Logic & Computer Organization (V23CST03)	Remove the Number System concept as it was already covered in BEEE(II SEM)	Removed
III	Advanced Data Structures & Algorithm Analysis (V23CST04)	• Move Dynamic Programming Concepts to UNIT-IV. • Move Branch and Bound to UNIT-V. • Remove Chromatic Number Decision Problem and Scheduling Identical Processors from UNIT-V. • Restrict Cook's theorem to without Proof.	Modified Accordingly
III	Object Oriented Programming Through Java (V23CST05)	Remove the JDBC Concepts from UNIT-V	Removed
III	Object Oriented Programming Through Java Lab (V23CSL05)	Remove the JDBC Experiments in the Lab Course.	Removed

The revised Syllabus is given in **Annexure-III**.

Item No. 5: Approval of Proposed Skill Enhancement Courses and Syllabi offered in III and IV Semesters for other branches under V23 Regulation.

The Approved Course Structure and Syllabi is given in **Annexure-IV**.



**Chairperson of BOS
(Dr.D Jaya Kumari)**

Annexure-I

S.No.	I – Semester						
	Course Code	Category	Name of the Course	L	T	P	C
1.	V23ENT01	BS&H	Communicative English	2	0	0	2
2.	V23PHT01	BS&H	Engineering Physics	3	0	0	3
3.	V23MAT01	BS&H	Linear Algebra & Calculus	3	0	0	3
4.	V23CMT01	Engineering Science	Basic Civil & Mechanical Engineering	3	0	0	3
5.	V23CST01	Engineering Science	Introduction to Programming	3	0	0	3
6.	V23ENL01	BS&H	Communicative English Lab	0	0	2	1
7.	V23PHL01	BS&H	Engineering Physics Lab	0	0	2	1
8.	V23MEL01	Engineering Science	Engineering Workshop	0	0	3	1.5
9.	V23CSL01	Engineering Science	Computer Programming Lab	0	0	3	1.5
10.	V23SPT02	BS&H	Health and wellness, Yoga and sports	-	-	1	0.5
Total				14	0	11	19.5

Total Contact Hours: 25

S.No.	II – Semester						
	Course Code	Category	Name of the Course	L	T	P	C
1.	V23CHT01	BS&H	Applied Chemistry	3	0	0	3
2.	V23MAT02	BS&H	Differential Equations & Vector Calculus	3	0	0	3
3.	V23EET01	Engineering Science	Basic Electrical and Electronics Engineering	3	0	0	3
4.	V23MET01	Engineering Science	Engineering Graphics	1	0	4	3
5.	V23CSL02	Engineering Science	IT Workshop	0	0	2	1
6.	V23CST02	Professional Core	Data Structures	3	0	0	3
7.	V23CHL01	BS&H	Applied Chemistry Lab	0	0	2	1
8.	V23EEL01	Engineering Science	Electrical and Electronics Engineering Workshop	0	0	3	1.5
9.	V23CSL03	Professional Core	Data Structures Lab	0	0	3	1.5
10.	V23SPT01	BS&H	NSS/NCC/Scouts & Guides/Community Service	-	-	1	0.5
Total				13	0	15	20.5

Total Contact Hours: 28

Annexure-II

S.No.	III – Semester						
	Course Code	Category	Course	L	T	P	C
1.		BS&H	Discrete Mathematics & Graph Theory	3	0	0	3
2.		Management Course- I	Managerial Economics and Financial Analysis	2	0	0	2
3.	V23CST03	Engineering Science	Digital Logic & Computer Organization	3	0	0	3
4.	V23CST04	Professional Core	Advanced Data Structures & Algorithm Analysis	3	0	0	3
5.	V23CST05	Professional Core	Object Oriented Programming Through Java	3	0	0	3
6.	V23CSL04	Professional Core	Advanced Data Structures and Algorithm Analysis Lab	0	0	3	1.5
7.	V23CSL05	Professional Core	Object Oriented Programming Through Java Lab	0	0	3	1.5
8.	V23CSSE01	Skill Enhancement Course	Python Programming Lab	0	1	2	2
9.		BS&H	Design Thinking & Innovation	1	0	2	2
Total				15	01	10	21

Total Contact Hours: 26

S.No.	IV – Semester						
	Course Code	Category	Course	L	T	P	C
1.		BS&H	Universal human values – understanding harmony and Ethical human conduct	2	1	0	3
2.		Engineering Science/Basic Science	Probability & Statistics	3	0	0	3
3.	V23CST06	Professional Core	Operating Systems	3	0	0	3
4.	V23CST07	Professional Core	Database Management Systems	3	0	0	3
5.	V23CST08	Professional Core	Software Engineering	2	1	0	3
6.	V23CSL06	Professional Core	Operating Systems Lab	0	0	3	1.5
7.	V23CSL07	Professional Core	Database Management Systems Lab	0	0	3	1.5
8.	V23CSSE02	Skill Enhancement Course	Full Stack Development –I	0	1	2	2
9.		Audit Course	Environmental Science	2	0	0	-
Total				15	03	08	20
Mandatory Community Service Project (V23COSP01) Internship of 08 weeks duration during summer vacation							

Total Contact Hours: 26

Annexure-III

Semester	III & IV	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CST03
Name of the Course	Digital Logic & Computer Organization					
Branch	Common to CSE, CST, CSE(AI) , AI&ML and CSE(DS)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Explain different Data Representation and various Combinational Digital Logic Circuits.(K2)

CO2: Explain various Sequential Digital Logic Circuits and basic Structure of Computers.(K2)

CO3: Describe the basic concepts of Computer Arithmetic and Processor Organization.(K2)

CO4: Illustrate different types of Memory and organization of Memory.(K2)

CO5: Demonstrate the ways of accessing various Interfacing devices with Processor.(K3)

UNIT-I:

Data Representation: Binary Numbers, Fixed Point Representation. Floating Point Representation. Signed binary numbers.

Digital Logic Circuits I: Basic Logic Functions, Minimization of Logic expressions. K-Map Simplification, Combinational Circuits, Decoders, Multiplexers.

UNIT-II:

Digital Logic Circuits-II: Sequential Circuits, Flip-Flops, Binary counters, Registers, Shift Registers, Ripple counters

Basic Structure of Computers: Computer Types, Functional units, Basic operational concepts, Bus structures, Software, Performance, multiprocessors and multi-computers, Computer Generations, Von-Neumann Architecture

UNIT-III:

Computer Arithmetic: Addition and Subtraction of Signed Numbers, Design of Fast Adders, Multiplication of Positive Numbers, Signed-operand Multiplication, Fast Multiplication, Integer Division, Floating-Point Numbers and Operations

Processor Organization: Fundamental Concepts, Execution of a Complete Instruction, Multiple-Bus Organization, Hardwired Control and Multi programmed Control

UNIT-IV:

The Memory Organization: Basic Concepts, Semiconductor RAM Memories, Read-Only Memories, Speed, Size and Cost, Cache Memories, Performance Considerations, Virtual Memories, Memory Management Requirements, Secondary Storage

UNIT-V:

Input / Output Organization: Accessing I/O Devices, Interrupts, Processor Examples, Direct Memory Access, Buses, Interface Circuits, Standard I/O Interfaces

Textbooks:

1. Computer Organization, Carl Hamacher, Zvonko Vranesic, Safwat Zaky, 6th edition, McGraw-Hill
2. Digital Design, 6th Edition, M. Morris Mano, Pearson Education.

Reference Books:

1. Computer Organization and Architecture, William Stallings, 11th Edition, Pearson.
2. Computer Systems Architecture, M. Moris Mano, 3rd Edition, Pearson
3. Computer Organization and Design, David A. Paterson, John L. Hennessy, Elsevier
4. Fundamentals of Logic Design, Roth, 5th Edition, Thomson

Online Learning Resources:

1. <https://nptel.ac.in/courses/106/103/106103068/>

Semester	III	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CST04
Name of the Course	Advanced Data Structures & Algorithm Analysis					
Branch	Common to CSE, CST, CSE(AI) , AI&ML and CSE(DS)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Demonstrate asymptotic notation and non-linear data structures like AVL and B-Trees.(K3)

CO2: Demonstrate graphs and divide and conquer technique.(K3)

CO3: Use greedy approach to determine various problems.(K3)

CO4: Apply Dynamic Programming and Backtracking techniques.(K3)

CO5: Solve different problems using Branch and Bound ,NP Hard and NP Complete.(K3)

UNIT-I

Introduction to Algorithm Analysis, Space and Time Complexity analysis, Asymptotic Notations.

AVL Trees – Creation, Insertion, Deletion operations and Applications.

B-Trees– Creation, Insertion, Deletion operations and Applications.

UNIT-II

Heap Trees (Priority Queues) – Min and Max Heaps, Operations and Applications.

Graphs – Terminology, Representations, Basic Search and Traversals, Connected Components and Biconnected Components, applications

Divide and Conquer: The General Method, Quick Sort, Merge Sort, Strassen's matrix multiplication, Convex Hull.

UNIT-III

Greedy Method: General Method, Job Sequencing with deadlines, Knapsack Problem, Minimum cost spanning trees, Single Source Shortest Paths– Dijkstra's .

UNIT-IV

Dynamic Programming: General Method, All pairs shortest paths, Single Source Shortest Paths–General Weights (Bellman Ford Algorithm), Optimal Binary Search Trees, 0/1 Knapsack, String Editing, Travelling Salesperson problem.

Backtracking: General Method, 8-Queens Problem, Sum of Subsets problem, Graph Coloring, 0/1 Knapsack Problem.

UNIT-V

Branch and Bound: The General Method, 0/1 Knapsack Problem, Travelling Salesperson problem.

NP Hard and NP Complete Problems: Basic Concepts, Cook's theorem (without Proof).

NP Hard Graph Problems: Clique Decision Problem (CDP), Traveling Salesperson Decision Problem (TSP).

NP Hard Scheduling Problems: Job Shop Scheduling.

Textbooks:

1. Fundamentals of Data Structures Horowitz, Ellis; Sahni, Sartaj; Mehta, Dinesh, 2nd Edition Universities Press

Reference Books:

1. Data Structures and program design in C, Robert Kruse, Pearson Education Asia
2. An introduction to Data Structures with applications, Trembley & Sorenson, McGraw Hill
3. The Art of Computer Programming, Vol.1: Fundamental Algorithms, Donald E Knuth, Addison-Wesley, 1997.
4. Data Structures using C & C++: Langsam, Augenstein & Tanenbaum, Pearson, 1995
5. Algorithms + Data Structures & Programs:, N. Wirth, PHI
6. Fundamentals of Data Structures in C++: Horowitz Sahni & Mehta, Galgotia Pub.
7. Data structures in Java:, Thomas Standish, Pearson Education Asia

Online Learning Resources:

1. https://www.tutorialspoint.com/advanced_data_structures/index.asp
2. <http://peterindia.net/Algorithms.html>
3. Abdul Bari, Introduction to Algorithms (youtube.com)

Semester	III	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CST05
Name of the Course	Object Oriented Programming Through Java					
Branch	Common to CSE, CST, CSE(AI) , AI&ML and CSE(DS)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1:Demonstrate the object-oriented programming principles with Java programming environment. **(K3)**

CO2:Demonstrate the concepts like classes, objects, argument passing mechanism, overloading and overriding. **(K3)**

CO3:Illustrate the concepts of arrays, inheritance and interfaces. **(K3)**

CO4: Demonstrate packages , java libraries, exception handling, java I/O and File concepts. **(K3)**

CO5:Illustrate the concepts of string handling, multithreading and Java FX GUI. **(K3)**

UNIT-I

Object Oriented Programming: Basic concepts, Principles, Program Structure in Java: Introduction, Writing Simple Java Programs, Elements or Tokens in Java Programs, Java Statements, Command Line Arguments, User Input to Programs, Escape Sequences Comments, Programming Style.

Data Types, Variables, and Operators: Introduction, Data Types in Java, Declaration of Variables, Data Types, Type Casting, Scope of Variable Identifier, Literal Constants, Symbolic Constants, Formatted Output with printf() Method, Static Variables and Methods, Attribute Final, Introduction to Operators, Precedence and Associativity of Operators, Assignment Operator (=), Basic Arithmetic Operators, Increment (++) and Decrement (- -) Operators, Ternary Operator, Relational Operators, Boolean Logical Operators, Bitwise Logical Operators.

Control Statements: Introduction, if Expression, Nested if Expressions, if-else Expressions, Ternary Operator?:, Switch Statement, Iteration Statements, while Expression, do-while Loop, for Loop, Nested for Loop, For-Each for Loop, Break Statement, Continue Statement.

UNIT-II

Classes and Objects: Introduction, Class Declaration and Modifiers, Class Members, Declaration of Class Objects, Assigning One Object to Another, Access Control for Class Members, Accessing Private Members of Class, Constructor Methods for Class, Overloaded Constructor Methods, Nested Classes, Final Class and Methods, Passing Arguments by Value and by Reference, Keyword this.

Methods: Introduction, Defining Methods, Overloaded Methods, Overloaded Constructor Methods, Class Objects as Parameters in Methods, Access Control, Recursive Methods, Nesting of Methods, Overriding Methods, Attributes Final and Static.

UNIT-III

Arrays: Introduction, Declaration and Initialization of Arrays, Storage of Array in Computer Memory, Accessing Elements of Arrays, Operations on Array Elements, Assigning Array to Another Array, Dynamic Change of Array Size, Sorting of Arrays, Search for Values in Arrays, Class Arrays, Two-dimensional Arrays, Arrays of Varying Lengths, Three- dimensional Arrays, Arrays as Vectors.

Inheritance: Introduction, Process of Inheritance, Types of Inheritances, Universal Super Class-Object Class, Inhibiting Inheritance of Class Using Final, Access Control and Inheritance, Multilevel Inheritance, Application of Keyword Super, Constructor Method and Inheritance, Method Overriding, Dynamic Method Dispatch, Abstract Classes, Interfaces and Inheritance.

Interfaces: Introduction, Declaration of Interface, Implementation of Interface, Multiple Interfaces, Nested Interfaces, Inheritance of Interfaces, Default Methods in Interfaces, Static Methods in Interface, Functional Interfaces, Annotations.

UNIT-IV

Packages and Java Library: Introduction, Defining Package, Importing Packages and Classes into Programs, Path and Class Path, Access Control, Packages in Java SE, Java.lang Package and its Classes, Class Object, Enumeration, class Math, Wrapper Classes, Auto- boxing and Auto-unboxing, Java util Classes and Interfaces, Formatter Class, Random Class, Time Package, Class Instant (java.time.Instant), Formatting for Date/Time in Java, Temporal Adjusters Class, Temporal Adjusters Class.

Exception Handling: Introduction, Hierarchy of Standard Exception Classes, Keywords throws and throw, try, catch, and finally Blocks, Multiple Catch Clauses, Class Throwable, Unchecked Exceptions, Checked Exceptions.

Java I/O and File: Java I/O API, standard I/O streams, types, Byte streams, Character streams, Scanner class, Files in Java(Text Book 2)

UNIT-V

String Handling in Java: Introduction, Interface Char Sequence, Class String, Methods for Extracting Characters from Strings, Comparison, Modifying, Searching; Class String Buffer.

Multithreaded Programming: Introduction, Need for Multiple Threads Multithreaded Programming for Multi-core Processor, Thread Class, Main Thread-Creation of New Threads, Thread States, Thread Priority-Synchronization, Deadlock and Race Situations, Inter-thread Communication - Suspending, Resuming, and Stopping of Threads.

Java FX GUI: Java FX Scene Builder, Java FX App Window Structure, displaying text and image, event handling, laying out nodes in scene graph, mouse events (Text Book 3)

Textbooks:

1. JAVA one step ahead, Anitha Seth, B.L.Juneja, Oxford.
2. Joy with JAVA, Fundamentals of Object Oriented Programming, DebasisSamanta, MonalisaSarma, Cambridge, 2023.
3. JAVA 9 for Programmers, Paul Deitel, Harvey Deitel, 4th Edition, Pearson.

Reference Books:

1. The complete Reference Java, 11th edition, Herbert Schildt, TMH
2. Introduction to Java programming, 7th Edition, Y Daniel Liang, Pearson

Online Learning Resources:

1. <https://nptel.ac.in/courses/106/105/106105191/>
2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_012880464547618816347_shared/overview

Semester	III	L	T	P	C	COURSE CODE
Regulation	V23	0	0	3	1.5	V23CSL04
Name of the Course	Advanced Data Structures & Algorithm Analysis Lab					
Branch	Common to CSE, CST, CSE(AI) , AI&ML and CSE(DS)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Demonstrate programs on AVL Trees and Heap trees. **(K3)**

CO2: Develop programs on Sorting algorithms and Graph traversal algorithms. **(K3)**

CO3: Develop programs using Greedy and Dynamic programming technique. **(K3)**

CO4: Develop programs using Backtracking and branch & Bound technique. **(K3)**

Experiments covering the Topics:

- Operations on AVL trees, Heap Trees
- Graph Traversals
- Sorting techniques
- Minimum cost spanning trees
- Shortest path algorithms
- 0/1 Knapsack Problem
- Travelling Salesperson problem
- Optimal Binary Search Trees
- N-Queens Problem
- Job Sequencing

Sample Programs:

1. Construct an AVL tree for a given set of elements which are stored in a file. And implement insert and delete operation on the constructed tree. Write contents of tree into a new file using in-order.
2. Construct B-Tree an order of 5 with a set of 100 random elements stored in array. Implement searching, insertion and deletion operations.
3. Construct Min and Max Heap using arrays, delete any element and display the content of the Heap.
4. Demonstrate BFT and DFT for given graph, when graph is represented by
 - a) Adjacency Matrix
 - b)Adjacency Lists
5. Develop a program for finding the biconnected components in a given graph.
6. Implement Quick sort and Merge sort and observe the execution time for various input sizes(Average, Worst and Best cases).
7. Examine the performance of Single Source Shortest Paths using Greedy method when the graph is represented by adjacency matrix and adjacency lists.
8. Demonstrate Job Sequencing with deadlines using Greedy strategy.
9. Demonstrate a program to solve 0/1Knapsack problem Using Dynamic Programming.
10. Demonstrate N-Queens Problem Using Backtracking.
11. Use Backtracking strategy to solve 0/1 Knapsack problem.
12. Demonstrate Travelling Sales Person problem using Branch and Bound approach.

Reference Books:

1. Fundamentals of Data Structures, Horowitz Ellis, SahniSartaj, Mehta, Dinesh, 2ndEdition, Universities Press
2. Computer Algorithms/C++ Ellis Horowitz, SartajSahni, SanguthevarRajasekaran, 2ndEdition, University Press
3. Data Structures and program design in C, Robert Kruse, Pearson Education Asia
4. An introduction to Data Structures with applications, Trembley& Sorenson, McGraw Hill

Online Learning Resources:

1. <http://cse01-iiith.vlabs.ac.in/>
2. <http://peterindia.net/Algorithms.html>

Semester	III	L	T	P	C	COURSE CODE
Regulation	V23	0	0	3	1.5	V23CSL05
Name of the Course	Object Oriented Programming Through Java Lab					
Branch	Common to CSE, CST, CSE(AI) , AI&ML and CSE(DS)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1:Construct programs to handle classes and objects. (K3)

CO2: Develop programs that incorporate inheritance and interfaces. (K3)

CO3:Construct programs on exception handling and File I/O. (K3)

CO4:Develop programs using multithreading and Java FX. (K3)

Experiments covering the Topics:

- Object Oriented Programming fundamentals- data types, control structures
- Classes, methods, objects, Inheritance, polymorphism,
- Exception handling, Threads, Packages, Interfaces
- Files, I/O streams, Java FX GUI

Exercise – 1:

- a) Develop a JAVA program to display default value of all primitive data type of JAVA
- b) Develop a java program that display the roots of a quadratic equation $ax^2+bx=0$. Find the discriminate D and basing on value of D, describe the nature of root.

Exercise – 2:

- a) Develop a JAVA program to search for an element in a given list of elements using binary search mechanism.
- b) Develop a JAVA program to sort for an element in a given list of elements using bubble sort
- c) Develop a JAVA program using String Buffer to delete, remove character.

Exercise – 3:

- a) Develop a JAVA program to implement class Mechanism. Build a class, methods and invoke them inside main method.
- b) Develop a JAVA program to implement method overloading.
- c) Develop a JAVA program to implement constructor.
- d) Develop a JAVA program to implement constructor overloading.

Exercise – 4:

- a) Develop a JAVA program to implement Single Inheritance.
- b) Develop a JAVA program to implement multi level Inheritance.
- c) Develop a JAVA program for abstract class to find areas of different shapes.

Exercise – 5:

- a) Develop a JAVA program on “super” keyword.
- b) Develop a JAVA program to implement Interface. What kind of Inheritance can be achieved?
- c) Develop a JAVA program that implements Runtime polymorphism.

Exercise – 6:

- a) Develop a JAVA program that describes exception handling mechanism.

- b) Develop a JAVA program on Multiple catch clauses.
- c) Develop a JAVA program to generate Built-in Exceptions.
- d) Develop a JAVA program to generate User Defined Exception.

Exercise – 7:

- a) Develop a JAVA program that creates threads by extending Thread class. First thread display “Good Morning “every 1 sec, the second thread displays “Hello “every 2 seconds and the third display “Welcome” every 3 seconds,(Repeat the same by implementing Runnable)
- b) Develop a program on isAlive() and join ().
- c) Demonstrate Daemon Threads.
- d) Demonstrate Producer Consumer Problem.

Exercise – 8:

- a) Develop a JAVA program that import and use the user defined packages.
- b) Demonstrate a GUI that display text in label and image in an ImageView (use JavaFX) Without writing any code.
- c) Develop a Tip Calculator app using several JavaFX components and learn how to respond to user interactions with the GUI

Textbooks:

1. JAVA one step ahead, Anitha Seth, B.L.Juneja, Oxford.
2. Joy with JAVA, Fundamentals of Object Oriented Programming, DebasisSamanta, MonalisaSarma, Cambridge, 2023.
3. JAVA 9 for Programmers, Paul Deitel, Harvey Deitel, 4th Edition, Pearson.

Reference Books:

1. The complete Reference Java, 11th edition, Herbert Schildt, TMH
2. Introduction to Java programming, 7th Edition, Y Daniel Liang, Pearson

Online Learning Resources:

1. <https://nptel.ac.in/courses/106/105/106105191/>
2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_012880464547618816347_shared/overview

Semester	III & IV	L	T	P	C	COURSE CODE
Regulation	V23	0	1	2	2	V23CSSE01
Name of the Course	Python Programming Lab					
Branch	Common to CSE, CST, CSE(AI) , AI&ML , CSE(DS) and EEE					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Illustrate basic concepts and control structures in python Programming. **(K3)**

CO2: Demonstrate functions and packages. **(K3)**

CO3: Construct python programs using structured data types. **(K3)**

CO4: Develop programs on Files, Exception handling and OOPs Concepts. **(K3)**

CO5: Construct programs for Data Analysis using Num Py and Pandas. **(K3)**

UNIT-I

History of Python Programming Language, Thrust Areas of Python, Installing Anaconda Python Distribution, Installing and Using Jupyter Notebook.

Parts of Python Programming Language: Identifiers, Keywords, Statements and Expressions, Variables, Operators, Precedence and Associativity, Data Types, Indentation, Comments, Reading Input, Print Output, Type Conversions, the type () Function and Is Operator, Dynamic and Strongly Typed Language.

Control Flow Statements: if statement, if-else statement, if...elif...else, Nested if statement, while Loop, for Loop, continue and break Statements, Catching Exceptions Using try and except Statement.

Sample Experiments:

1. Develop a program to find the largest element among three Numbers.
2. Develop a Program to display all prime numbers within an interval
3. Develop a program to swap two numbers without using a temporary variable.
4. Demonstrate the following Operators in Python with suitable examples.
 - i) Arithmetic Operators
 - ii) Relational Operators
 - iii) Assignment Operators
 - iv) Logical Operators
 - v) Bit wise Operators
 - vi) Ternary Operator
 - vii) Membership Operators
 - viii) Identity Operators
5. Develop a program to add and multiply complex numbers
6. Develop a program to print multiplication table of a given number.

UNIT-II

Functions: Built-In Functions, Commonly Used Modules, Function Definition and Calling the function, return Statement and void Function, Scope and Lifetime of Variables, Default Parameters, Keyword Arguments, *args and **kwargs, Command Line Arguments.

Strings: Creating and Storing Strings, Basic String Operations, Accessing Characters in String by Index Number, String Slicing and Joining, String Methods, Formatting Strings.

Lists: Creating Lists, Basic List Operations, Indexing and Slicing in Lists, Built-In Functions Used on Lists, List Methods, del Statement.

Sample Experiments:

1. Develop a program to define a function with multiple return values.
2. Develop a program to define a function using default arguments.
3. Develop a program to find the length of the string without using any library functions.

4. Develop a program to check if the substring is present in a given string or not.
5. Develop a program to perform the given operations on a list:
 - i. addition ii. Insertion iii. slicing
6. Develop a program to perform any 5 built-in functions by taking any list.

UNIT-III

Dictionaries: Creating Dictionary, Accessing and Modifying key: value Pairs in Dictionaries, Built-In Functions Used on Dictionaries, Dictionary Methods, del Statement.

Tuples and Sets: Creating Tuples, Basic Tuple Operations, tuple() Function, Indexing and Slicing in Tuples, Built-In Functions Used on Tuples, Relation between Tuples and Lists, Relation between Tuples and Dictionaries, Using zip() Function, Sets, Set Methods, Frozenset.

Sample Experiments:

1. Develop a program to create tuples (name, age, address, college) for at least two members and concatenate the tuples and print the concatenated tuples.
2. Develop a program to count the number of vowels in a string (No control flow allowed).
3. Develop a program to check if a given key exists in a dictionary or not.
4. Develop a program to add a new key-value pair to an existing dictionary.
5. Develop a program to sum all the items in a given dictionary.

UNIT-IV

Files: Types of Files, Creating and Reading Text Data, File Methods to Read and Write Data, Reading and Writing Binary Files, Pickle Module, Reading and Writing CSV Files, Python os and os.path Modules.

Object-Oriented Programming: Classes and Objects, Creating Classes in Python, Creating Objects in Python, Constructor Method, Classes with Multiple Objects, Class Attributes Vs Data Attributes, Encapsulation, Inheritance, Polymorphism.

Sample Experiments:

1. Develop a program to sort words in a file and put them in another file. The output file should have only lower-case words, so any upper-case words from source must be lowered.
2. Develop a Python program to print each line of a file in reverse order.
3. Develop a Python program to compute the number of characters, words and lines in a file.
4. Develop a program to create, display, append, insert and reverse the order of the items in the array.
5. Develop a program to add, transpose and multiply two matrices.
6. Develop a Python program to create a class that represents a shape. Include methods to calculate its area and perimeter. Implement subclasses for different shapes like circle, triangle, and square.

UNIT-V

Introduction to Data Science: Functional Programming, JSON and XML in Python, NumPy with Python, Pandas.

Sample Experiments:

1. Develop a Python program to check whether a JSON string contains complex object or not.
2. Demonstrate NumPy arrays creation using array () function.
3. Demonstrate use of ndim, shape, size, dtype.
4. Demonstrate basic slicing, integer and Boolean indexing.
5. Develop a Python program to find min, max, sum, cumulative sum of array
6. Prepare a dictionary with at least five keys and each key represent value as a list where this list contains at least ten values and convert this dictionary as a pandas data frame and explore the data through the data

frame as follows:

- a) Apply head () function to the pandas data frame
 - b) Perform various data selection operations on Data Frame
7. Select any two columns from the above data frame, and observe the change in one attribute with respect to other attribute with scatter and plot operations in matplotlib.

Reference Books:

1. Gowrishankar S, Veena A., Introduction to Python Programming, CRC Press.
2. Python Programming, S Sridhar, J Indumathi, V M Hariharan, 2ndEdition, Pearson, 2024
3. Introduction to Programming Using Python, Y. Daniel Liang, Pearson.

Online Learning Resources:

1. <https://www.coursera.org/learn/python-for-applied-data-science-ai>
2. <https://www.coursera.org/learn/python?specialization=python#syllabus>

Semester	IV	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CST06
Name of the Course	Operating Systems					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Describe Operating System Services, System Calls, Building, Booting and Debugging. **(K2)**

CO2: Illustrate Process Management Concepts and CPU Scheduling Algorithms. **(K3)**

CO3: Demonstrate Process Synchronization primitives and Deadlock handling techniques. **(K3)**

CO4: Illustrate Memory Management Techniques and Page Replacement Algorithms. **(K3)**

CO5: Describe File System Concepts and Mass Storage Structures. **(K2)**

UNIT-I

Operating Systems Overview: Introduction, Operating system functions, Operating systems operations, Computing environments, Free and Open-Source Operating Systems

System Structures: Operating System Services, User and Operating-System Interface, system calls, Types of System Calls, system programs, Operating system Design and Implementation, Operating system structure, Building and Booting an Operating System, Operating system debugging

UNIT-II

Processes: Process Concept, Process scheduling, Operations on processes, Inter-process communication.

Threads and Concurrency: Multithreading models, Thread libraries, Threading issues.

CPU Scheduling: Basic concepts, Scheduling criteria, Scheduling algorithms, Multiple processor scheduling.

UNIT-III

Synchronization Tools: The Critical Section Problem, Peterson's Solution, Mutex Locks, Semaphores, Monitors, Classic problems of Synchronization.

Deadlocks: system Model, Deadlock characterization, Methods for handling Deadlocks, Deadlock prevention, Deadlock avoidance, Deadlock detection, Recovery from Deadlock.

UNIT-IV

Memory-Management Strategies: Introduction, Contiguous memory allocation, Paging, Structure of the Page Table, Swapping.

Virtual Memory Management: Introduction, Demand paging, Copy-on-write, Page replacement, Allocation of frames, Thrashing

Storage Management: Overview of Mass Storage Structure, HDD Scheduling.

UNIT-V

File System: File System Interface: File concept, Access methods, Directory Structure; File system

Implementation: File-system structure, File-system Operations, Directory

implementation, Allocation method, Free space management; File-System Internals: File-System Mounting, Partitions and Mounting, File Sharing.

Protection: Goals of protection, Principles of protection, Protection Rings, Domain of protection, Access matrix.

Textbooks:

1. Operating System Concepts, Silberschatz A, Galvin P B, Gagne G, 10th Edition, Wiley, 2018.
2. Modern Operating Systems, Tanenbaum A S, 4th Edition, Pearson , 2016

Reference Books:

1. Operating Systems -Internals and Design Principles, Stallings W, 9th edition, Pearson, 2018
2. Operating Systems: A Concept Based Approach, D.M Dhamdhere, 3rd Edition, McGraw- Hill, 2013

Online Learning Resources:

1. <https://nptel.ac.in/courses/106/106/106106144/>
2. <http://peterindia.net/OperatingSystems.html>

Semester	IV	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CST07
Name of the Course	Database Management Systems					
Branch	Common to CSE, CST, CSE(AI) , AI&ML and CSE(DS)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Describe Database systems, various Data models and Database architecture. **(K2)**

CO2: Develop various real time applications using Relational algebra and Relational calculus. **(K3)**

CO3: Illustrate Nested queries and different types of joins. **(K3)**

CO4: Apply various Normalization techniques to refine schema. **(K3)**

CO5: Explain Transaction management, Concurrency control and Database indexing techniques. **(K2)**

UNIT-I

Introduction: Database system, Characteristics (Database Vs File System), Database Users, Advantages of Database systems, Database applications. Brief introduction of different Data Models; Concepts of Schema, Instance and data independence; Three tier schema architecture for data independence; Database system structure, environment, Centralized and Client Server architecture for the database.

Entity Relationship Model: Introduction, Representation of entities, attributes, entity set, relationship, relationship set, constraints, sub classes, super class, inheritance, specialization, generalization using ER Diagrams.

UNIT-II

Relational Model: Introduction to relational model, concepts of domain, attribute, tuple, relation, importance of null values, constraints (Domain, Key constraints, integrity constraints) and their importance, Relational Algebra, Relational Calculus.

BASIC SQL: Simple Database schema, data types, table definitions (create, alter), different DML operations (insert, delete, update).

UNIT-III

SQL: Basic SQL querying (select and project) using where clause, arithmetic & logical operations, SQL functions(Date and Time, Numeric, String conversion).Creating tables with relationship, implementation of key and integrity constraints, nested queries, sub queries, grouping, aggregation, ordering, implementation of different types of joins, view(updatable and non-updatable), relational set operations.

UNIT-IV

Schema Refinement (Normalization): Purpose of Normalization or schema refinement, concept of functional dependency, normal forms based on functional dependency Lossless join and dependency preserving decomposition, (1NF, 2NF and 3 NF), concept of surrogate key, Boyce-Codd normal form(BCNF), MVD, Fourth normal form(4NF), Fifth Normal Form (5NF).

UNIT-V

Transaction Concept: Transaction State, ACID properties, Concurrent Executions, Serializability, Recoverability, Implementation of Isolation, Testing for Serializability, lock based, time stamp based, optimistic, concurrency

protocols, Deadlocks, Failure Classification, Storage, Recovery and Atomicity, Recovery algorithm.

Introduction to Indexing Techniques: B+ Trees, operations on B+Trees, Hash Based Indexing.

Textbooks:

1. Database Management Systems, 3rd edition, Raghurama Krishnan, Johannes Gehrke, TMH (For Chapters 2, 3, 4)
2. Database System Concepts, 5th edition, Silberschatz, Korth, Sudarsan, TMH (For Chapter 1 and Chapter 5)

Reference Books:

1. Introduction to Database Systems, 8th edition, C J Date, Pearson.
2. Database Management System, 6th edition, Ramez Elmasri, Shamkant B. Navathe, Pearson
3. Database Principles Fundamentals of Design Implementation and Management, Corlos Coronel, Steven Morris, Peter Robb, Cengage Learning.

Online Learning Resources:

1. <https://nptel.ac.in/courses/106/105/106105175/>
2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01275806667282022456_shared/overview.

Semester	IV	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CST08
Name of the Course	Software Engineering					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Demonstrate the evolution, development practices, and various software life cycle models. **(K3)**

CO2: Illustrate project management techniques and explain requirements analysis and specification. **(K3)**

CO3: Develop effective software designs, using agile methodologies and user interface principles. **(K3)**

CO4: Apply coding practices, conduct thorough testing, and manage software quality. **(K3)**

CO5: Distinguish the use of CASE tools, manage software maintenance, and implement software reuse strategies. **(K4)**

UNIT-I

Introduction: Evolution, Software development projects, Exploratory style of software developments, Emergence of software engineering, Notable changes in software development practices, Computer system engineering.

Software Life Cycle Models: Basic concepts, Waterfall model and its extensions, Rapid application development, Agile development model, Spiral model.

UNIT-II

Software Project Management: Software project management complexities, Responsibilities of a software project manager, Metrics for project size estimation, Project estimation techniques, Empirical Estimation techniques, COCOMO, Halstead's software science, risk management.

Requirements Analysis And Specification: Requirements gathering and analysis, Software Requirements Specification (SRS), Formal system specification, Axiomatic specification, Algebraic specification, Executable specification and 4GL.

UNIT-III

Software Design: Overview of the design process, How to characterize a good software design? Layered arrangement of modules, Cohesion and Coupling. approaches to software design.

Agility: Agility and the Cost of Change, Agile Process, Extreme Programming (XP), Other Agile Process Models, Tool Set for the Agile Process (Text Book 2)

Function-Oriented Software Design: Overview of SA/SD methodology, Structured analysis, Developing the DFD model of a system, Structured design, Detailed design, and Design Review.

User Interface Design: Characteristics of a good user interface, Basic concepts, Types of user interfaces, Fundamentals of component-based GUI development, and user interface design methodology.

UNIT-IV

Coding And Testing: Coding, Code review, Software documentation, Testing, Black-box testing, White-Box testing, Debugging, Program analysis tools, Integration testing, Testing object-oriented programs, Smoke testing, and Some general issues associated with testing.

Software Reliability and Quality Management: Software reliability. Statistical testing, Software quality, Software quality management system, ISO 9000.SEI Capability maturity model. Few other important quality standards, and Six Sigma.

UNIT-V

Computer-Aided Software Engineering (Case): CASE and its scope, CASE environment, CASE support in the software life cycle, other characteristics of CASE tools, Towards second generation CASE Tool, and Architecture of a CASE Environment.

Software Maintenance: Characteristics of software maintenance, Software reverse engineering, Software maintenance process models and Estimation of maintenance cost.

Software Reuse: reuse- definition, introduction, reason behind no reuse so far, Basic issues in any reuse program, A reuse approach, and Reuse at organization level.

Textbooks:

1. Fundamentals of Software Engineering, Rajib Mall, 5th Edition, PHI.
2. Software Engineering A practitioner's Approach, Roger S. Pressman, 9th Edition, Mc- Graw Hill International Edition.

Reference Books:

1. Software Engineering, Ian Sommerville, 10th Edition, Pearson.
2. Software Engineering, Principles and Practices, Deepak Jain, Oxford University Press.

Online Learning Resources:

1. <https://nptel.ac.in/courses/106/105/106105182/>
2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01260589506387148827_shared/overview
3. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013382690411003904735_shared/overview

Semester	IV	L	T	P	C	COURSE CODE
Regulation	V23	0	0	3	1.5	V23CSL06
Name of the Course	Operating Systems Lab					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Practice of UNIX fundamentals, commands & system calls. **(K3)**

CO2: Construct programs using CPU scheduling algorithms. **(K3)**

CO3: Demonstrate IPC, semaphores, monitors, deadlocks. **(K3)**

CO4: Develop programs to solve Page replacement algorithms, file allocation strategies and memory allocation strategies. **(K3)**

Experiments covering the Topics:

- UNIX fundamentals, commands & system calls
- CPU Scheduling algorithms, thread processing
- IPC, semaphores, monitors, deadlocks
- Page replacement algorithms, file allocation strategies
- Memory allocation strategies

Sample Programs:

1. Practice Basic UNIX Commands.
2. Develop programs using the following UNIX operating system calls fork, exec, getpid, exit, wait, close, stat, opendir and readdir
3. Demonstrate UNIX commands like cp, ls, grep, etc.,
4. Demonstrate the following CPU scheduling algorithms
a) FCFS b) SJF c) Priority d) Round Robin
5. Develop a program to control the number of ports opened by the operating system with
a) Semaphore b) Monitors.
6. Illustrate concurrent execution of threads using pthreads library.
7. Solve producer-consumer problem using Semaphores.
8. Demonstrate the following memory allocation methods for fixed partition
a) First fit b) Worst fit c) Best fit
9. Develop programs to simulate the following page replacement algorithms
a) FIFO b) LRU c) LFU
10. Develop program to simulate Paging Technique of memory management.
11. Demonstrate Bankers Algorithm for Dead Lock avoidance and prevention.
12. Develop program to simulate the following file allocation strategies
a) Sequential b) Indexed c) Linked
13. Download and install nachos operating system and experiment with it.

Reference Books:

1. Operating System Concepts, Silberschatz A, Galvin P B, Gagne G, 10th Edition, Wiley, 2018.
2. Modern Operating Systems, Tanenbaum A S, 4th Edition, Pearson, 2016
3. Operating Systems -Internals and Design Principles, Stallings W, 9th edition, Pearson, 2018.
4. Operating Systems: A Concept Based Approach, D.M Dhamdhare, 3rd Edition, McGraw- Hill, 2013

Online Learning Resources:

1. <https://www.cse.iitb.ac.in/~mythili/os/>
2. <http://peterindia.net/OperatingSystems.html>
3. www.cs.washington.edu/~tom/nachos

Semester	IV	L	T	P	C	COURSE CODE
Regulation	V23	0	0	3	1.5	V23CSL07
Name of the Course	Database Management Systems Lab					
Branch	Common to CSE, CST, CSE(AI) , AI&ML and CSE(DS)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Construct SQL queries using DDL,DML and DCL commands. **(K3)**

CO2: Experiment with nested queries, built-in functions and various constraints. **(K3)**

CO3: Construct PL/SQL programs on Control structures, Cursors and Triggers . **(K3)**

CO4: Develop a Java program to connect with a database using JDBC. **(K3)**

Experiments covering the Topics:

- DDL, DML, DCL commands
- Queries, nested queries, built-in functions,
- PL/SQL programming- control structures
- Procedures, Functions, Cursors, Triggers,
- Database connectivity- ODBC/JDBC

Sample Programs:

1. Demonstrate Creation, altering and dropping of tables and inserting rows into a table (use constraints while creating tables) examples using SELECT command.
2. Construct queries (along with sub queries) using ANY, ALL, IN, EXISTS, NOTEXISTS, UNION, INTERSET, Constraints. Example:- Select the roll number and name of the student who secured fourth rank in the class.
3. Construct queries using Aggregate functions (COUNT, SUM, AVG, MAX and MIN), GROUP BY, HAVING and Creation and dropping of Views.
4. Construct queries using Conversion functions (to_char, to_number and to_date), string functions (Concatenation, lpad, rpad, ltrim, rtrim, lower, upper, initcap, length, substr and instr), date functions (Sysdate, next_day, add_months, last_day, months_between, least, greatest, trunc, round, to_char, to_date)
5.
 - i. Develop a simple PL/SQL program which includes declaration section, executable section and exception –Handling section (Ex. Student marks can be selected from the table and printed for those who secured first class and an exception can be raised if no records were found)
 - ii. Insert data into student table and use COMMIT, ROLLBACK and SAVEPOINT in PL/SQL block.
6. Develop a program that includes the features NESTED IF, CASE and CASE expression. The program can be extended using the NULLIF and COALESCE functions.
7. Develop programs using WHILE LOOPS, numeric FOR LOOPS, nested loops using ERROR Handling, BUILT –IN Exceptions, USE defined Exceptions, RAISE- APPLICATION ERROR.
8. Develop programs using procedures, passing parameters IN and OUT of PROCEDURES.
9. Develop programs on stored functions, invoke functions in SQL Statements and complex functions.
10. Develop programs using features parameters in a CURSOR, FOR UPDATE CURSOR, WHERE CURRENT of clause and CURSOR variables.
11. Develop Programs using BEFORE and AFTER Triggers, Row and Statement Triggers and INSTEAD OF Triggers
12. Construct a table and perform the search operation on table using indexing and non- indexing techniques.

13. Develop a Java program that connects to a database using JDBC
14. Develop a Java program to connect to a database using JDBC and insert values into it
15. Develop a Java program to connect to a database using JDBC and delete values from it

Textbooks:

1. Database Management Systems, 3rd edition, Raghurama Krishnan, Johannes Gehrke, TMH (For Chapters 2, 3, 4)
2. Database System Concepts, 5th edition, Silberschatz, Korth, Sudarsan, TMH (For Chapter 1 and Chapter 5)

Reference Books:

1. Oracle: The Complete Reference by Oracle Press
2. Nilesh Shah, "Database Systems Using Oracle", PHI, 2007
3. Rick F Vander Lans, "Introduction to SQL", Fourth Edition, Pearson Education, 2007

Semester	IV	L	T	P	C	COURSE CODE
Regulation	V23	0	1	2	2	V23CSSE02
Name of the Course	FULL STACK DEVELOPMENT – I					
Branch	Common to CSE, CST, CSE(AI) and AI&ML					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Illustrate HTML elements and their attributes for designing static web pages. **(K3)**

CO2: Apply appropriate CSS styles to HTML elements. **(K3)**

CO3: Demonstrate JavaScript Pre-defined and User-defined Objects. **(K3)**

CO4: Develop dynamic web pages and validate forms using JavaScript. **(K3)**

Experiments covering the Topics:

- Lists, Links and Images
- HTML Tables, Forms and Frames
- HTML 5 and Cascading Style Sheets, Types of CSS
- Selector forms
- CSS with Color, Background, Font, Text and CSS Box Model
- Applying JavaScript - internal and external, I/O, Type Conversion
- JavaScript Conditional Statements and Loops, Pre-defined and User-defined Objects
- JavaScript Functions and Events
- Node.js

Sample Programs:

1. Lists, Links and Images

- a) Develop a HTML program, to explain the working of lists.
- b) Note: It should have an ordered list, unordered list, nested lists and ordered list in an unordered list and definition lists.
- c) Develop a HTML program, to explain the working of hyperlinks using <a> tag and href, target Attributes.
- d) Develop a HTML document that has your image and your friend's image with a specific height and width. Also when clicked on the images it should navigate to their respective profiles.
- e) Develop a HTML program, in such a way that, rather than placing large images on a page, the preferred technique is to use thumbnails by setting the height and width parameters to something like to 100*100 pixels. Each thumbnail image is also a link to a full sized version of the image. Create an image gallery using this technique

2. HTML Tables, Forms and Frames

- a) Develop a HTML program, to explain the working of tables. (use tags: <table>, <tr>, <th>,<td> and attributes: border, rowspan, colspan)
- b) Develop a HTML program, to explain the working of tables by preparing a timetable. (Note: Use <caption> tag to set the caption to the table & also use cell spacing, cell padding, border, rowspan, colspan etc.).
- c) Develop a HTML program, to explain the working of forms by designing Registration form. (Note: Include text field, password field, number field, date of birth field, checkboxes, radio buttons, list boxes

using <select>&<option> tags, <text area> and two buttons ie: submit and reset. Use tables to provide a better view).

- d) Develop a HTML program, to explain the working of frames, such that page is to be divided into 3 parts on either direction. (Note: first frame □ image, second frame □ paragraph, third frame □ hyperlink. And also make sure of using “no frame” attribute such that frames to be fixed).

3. HTML 5 and Cascading Style Sheets, Types of CSS

- a) Develop a HTML program, that makes use of <article>, <aside>, <figure>, <figcaption>,<footer>, <header>, <main>, <nav>, <section>, <div>, tags.
- b) Develop a HTML program, to embed audio and video into HTML web page.
- c) Develop a program to apply different types (or levels of styles or style specification formats)- inline, internal, external styles to HTML elements. (identify selector, property and value).

4. Selector forms

- a) Develop a program to apply different types of selector forms
 - i. Simple selector (element, id, class, group, universal)
 - ii. Combinator selector (descendant, child, adjacent sibling, general sibling)
 - iii. Pseudo-class selector
 - iv. Pseudo-element selector
 - v. Attribute selector

5. CSS with Color, Background, Font, Text and CSS Box Model

- a) Develop a program to demonstrate the various ways you can reference a color in CSS.
- b) Develop a CSS rule that places a background image halfway down the page, tilting it horizontally. The image should remain in place when the user scrolls up or down.
- c) Develop a program using the following terms related to CSS font and text:
 - i. font-size ii. font-weight iii. font-style
 - iv. text-decoration v. text-transformation vi. text-alignment
- d) Develop a program, to explain the importance of CSS Box model using
 - i. Content ii. Border iii. Margin iv. padding

6. Applying JavaScript - internal and external, I/O, Type Conversion

- a) Develop a program to embed internal and external JavaScript in a web page.
- b) Develop a program to explain the different ways for displaying output.
- c) Develop a program to explain the different ways for taking input.
- d) Create a webpage which uses prompt dialogue box to ask a voter for his name and age. Display the information in table format along with either the voter can vote or not

7. Java Script Pre-defined and User-defined Objects

- a) Develop a program using document object properties and methods.
- b) Develop a program using window object properties and methods.
- c) Develop a program using array object properties and methods.
- d) Develop a program using math object properties and methods.
- e) Develop a program using string object properties and methods.
- f) Develop a program using regex object properties and methods.
- g) Develop a program using date object properties and methods.
- h) Develop a program to explain user-defined object by using properties, methods, accessors, constructors and display.

8. Java Script Conditional Statements and Loops

- a) Develop a program which asks the user to enter three integers, obtains the numbers from the user and

outputs HTML text that displays the larger number followed by the words “LARGER NUMBER” in an information message dialog. If the numbers are equal, output HTML text as “EQUAL NUMBERS”.

- b) Develop a program to display week days using switch case.
- c) Develop a program to print 1 to 10 numbers using for, while and do-while loops.
- d) Develop a program to print data in object using for-in, for-each and for-of loops
- e) Develop a program to determine whether a given number is an ‘ARMSTRONG NUMBER’ or not.
[Eg: 153 is an Armstrong number, since sum of the cube of the digits is equal to the number i.e., $1^3 + 5^3 + 3^3 = 153$]
- f) Develop a program to display the denomination of the amount deposited in the bank in terms of 100’s, 50’s, 20’s, 10’s, 5’s, 2’s & 1’s. (Eg: If deposited amount is Rs.163, the output should be 1-100’s, 1-50’s, 1- 10’s, 1-2’s & 1-1’s)

9. Java Script Functions and Events

- a) Design a appropriate function should be called to display
 - i. Factorial of that number
 - ii. Fibonacci series up to that number
 - iii. Prime numbers up to that number
 - iv. Is it palindrome or not
- b) Prepare a HTML file having a text box and four buttons named Factorial, Fibonacci, Prime, and Palindrome. When a button is pressed an appropriate function should be called to display
 - i. Factorial of that number
 - ii. Fibonacci series up to that number
 - iii. Prime numbers up to that number
 - iv. Is it palindrome or not
- c) Develop a program to validate the following fields in a registration page
 - i. Name (start with alphabet and followed by alphanumeric and the length should not be less than 6 characters)
 - ii. Mobile (only numbers and length 10 digits)
 - iii. E-mail (should contain format like xxxxxxxx@xxxxxx.xxx)

Reference Books:

1. Programming the World Wide Web, 7th Edition, Robert W Sebesta, Pearson, 2013.
2. Web Programming with HTML5, CSS and JavaScript, John Dean, Jones & Bartlett Learning, 2019 (Chapters 1-11).
3. Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, Vasan Subramanian, 2nd edition, APress, O’Reilly.

Online Learning Resources:

1. <https://www.w3schools.com/html>
2. <https://www.w3schools.com/css>
3. <https://www.w3schools.com/js/>
4. <https://www.w3schools.com/nodejs>

Annexure-IV

Approved Courses and Syllabi for other Branches under V23 Regulation

S.No.	Category	Branch	SEM	Course Code	Course Name
1	Skill Enhancement Course	ECE	III	V23CSSE11	Data Structures using Python Lab
2		EEE	III	V23CSSE12	Data Structures Lab
3		MECH	III	V23CSSE13	Python Programming Essentials Lab

Semester	III	L	T	P	C	COURSE CODE
Regulation	V23	0	1	2	2	V23CSSE11
Name of the Course	Data Structures using Python Lab					
Branch	Common to ECE & ECT					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Apply Object-Oriented Programming (OOP) Principles in Python such as classes, objects, inheritance, polymorphism, encapsulation, and abstraction to solve real-world problems. **(K3)**

CO2: Apply Python Comprehensions such as list comprehensions, dictionary comprehensions, and set comprehensions and generator comprehensions to create and manipulate collections efficiently in Python programs. **(K3)**

CO3: Apply Searching and Sorting Techniques in Python such as linear search, binary search, bubble sort, selection sort, merge sort, and quick sort algorithms to efficiently organize and retrieve data in Python programs. **(K3)**

CO4: Develop Python programs to implement stacks, queues, and linked lists. **(K3)**

CO5: Construct binary search trees to efficiently store and retrieve data in Python programs. **(K3)**

List of Experiments:

1. Develop a Python program for class, Flower, that has three instance variables of type str, int, and float that respectively represent the name of the flower, its number of petals, and its price. Your class must include a constructor method that initializes each variable to an appropriate value, and your class should include methods for setting the value of each type, and retrieving the value of each type.
2. Develop an inheritance hierarchy based upon a Polygon class that has abstract methods area () and perimeter(). Implement classes Triangle, Quadrilateral, Pentagon, that extend this base class, with the obvious meanings for the area() and perimeter() methods. Write a simple program that allows users to create polygons of the various types and input their geometric dimensions, and the program then outputs their area and perimeter
3. Develop a python program to implement Method Overloading and Method Overriding.
4. Develop a Python program to illustrate the following comprehensions: a) List Comprehensions b) Dictionary Comprehensions c) Set Comprehensions d) Generator Comprehensions
5. Develop a Python program to generate the combinations of n distinct objects taken from the elements of a given list. Example: Original list: [1, 2, 3, 4, 5, 6, 7, 8, 9]
Combinations of 2 distinct objects: [1, 2] [1, 3] [1, 4] [1, 5] [7, 8] [7, 9] [8, 9].
6. Develop a program for Linear Search and Binary search.
7. Develop a program to implement Bubble Sort and Selection Sort.
8. Develop a program to implement Merge sort and Quick sort.
9. Develop a program to implement Stacks and Queues.
10. Develop a program to implement Singly Linked List.
11. Develop a program to implement Doubly Linked list.
12. Develop a program to implement Binary Search Tree.

Semester	III	L	T	P	C	COURSE CODE
Regulation	V23	0	1	2	2	V23CSSE12
Name of the Course	Data Structures Lab					
Branch	EEE					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Solve the problems such as sorting, searching, matrix multiplication by applying the concepts 1D and 2D arrays. **(K3)**

CO2: Construct efficient linked list data structure for basic operations and to solve real world problems such as addition of polynomials etc. **(K3)**

CO3: Construct efficient stack data structure using arrays and linked lists and its applications like infix to postfix conversion, evaluation of postfix expression, reversing a linked list etc. **.(K3)**

CO4: Construct efficient queue, circular queue, and dequeue data structures using arrays and linked lists. **(K3)**

CO5: Construct programs on basic operations and traversal techniques of binary search trees. **(K3)**

UNIT I

Introduction to Data Structures: Definition and importance of Data structures, Abstract data types (ADTs) and its specifications, Arrays: Introduction, 1-D, 2-D Arrays, accessing elements of array, Row Major and Column Major storage of Arrays, Searching Techniques: Linear & Binary Search, Sorting Techniques: Bubble sort, Selection sort, Quick sort.

Sample experiments:

1. Develop a Program to find min & max element in an array.
2. Develop a Program to implement matrix multiplication.
3. Develop a program Find an element in given list of sorted elements in an array using Binary search.
4. Develop a program to implement Selection and Quick sort techniques.

UNIT II

Linked Lists: Singly linked lists: representation and operations, doubly linked lists and circular linked lists, Comparing arrays and linked lists, Applications of linked lists.

1. Develop a program to implement the following operations.
a. Insert b. Deletion c. Traversal
2. Develop a program to store name, roll no, and marks of students in a class using circular double linked list.
3. Develop a program to perform addition of given two polynomial expressions using linked list.

UNIT III

Stacks: Introduction to stacks: properties and operations, implementing stacks using arrays and linked lists, Applications of stacks in expression evaluation, backtracking, reversing list etc.

Sample experiments:

1. Develop a program to implement stack operations using
 - a. Arrays
 - b. Linked list
2. Develop a program to convert given infix expression into postfix expression using stacks.
3. Develop a program to evaluate given postfix expression using stack.
4. Develop a program to reverse given linked list using stack.

UNIT IV

Queues: Introduction to queues: properties and operations, Circular queues, implementing queues using arrays and linked lists, Applications of queues scheduling, etc.

Deque: Introduction to deque (double-ended queues), Operations on deque and their applications.

Sample experiments:

1. Develop a program to implement Queue operations using
 - a. Arrays
 - b. Linked list
2. Develop a program to implement Circular Queue using
 - a. Arrays
 - b. Linked list
3. Develop a program to implement Dequeue using linked list.

UNIT V

Trees: Introduction to Trees, Binary trees and traversals, Binary Search Tree–Insertion, Deletion & Traversal

Sample experiments:

1. Develop a program to implement binary tree traversals using linked list.
2. Develop a program to create binary search tree for given list of integers. Perform in-order traversal of the tree. Implement insertion and deletion operations.

Textbooks:

1. Data Structures and algorithm analysis in C, Mark Allen Weiss, Pearson, 2nd Edition.
2. Fundamentals of data structures in C, Ellis Horowitz, Sartaj Sahni, Susan Anderson Freed, Silicon Press, 2008.

Reference Books:

1. Algorithms and Data Structures: The Basic Toolbox by Kurt Mehlhorn and Peter Sanders.
2. C Data Structures and Algorithms by Alfred V. Aho, Jeffrey D. Ullman, and John E. Hopcroft.
3. Problem Solving with Algorithms and Data Structures by Brad Miller and David Ranum.
4. Introduction to Algorithms by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein.
5. Algorithms in C, Parts 1- 5 (Bundle): Fundamentals, Data Structures, Sorting, Searching, and Graph Algorithms" by Robert Sedgewick.

Semester	III	L	T	P	C	COURSE CODE
Regulation	V23	0	1	2	2	V23CSSE13
Name of the Course	Python Programming Essentials Lab					
Branch	ME					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Develop basic python programs using control structures, functions, and modules. **(K3)**

CO2: Construct python programs using the data structures like lists, tuples, dictionaries, and sets. **(K3)**

CO3: Develop python programs using strings, file i/o, exception handling, and object-oriented programming (OOP) concepts. **(K3)**

CO4: Develop python programs using libraries numpy, pandas, matplotlib, seaborn. **(K3)**

CO5: Construct python programs to extract data from the web and work with databases using SQLite. **(K3)**

Experiment 1: Introduction to Python

- Objective: Install Python and setup the development environment.
- Tasks:
 - Install Python and an IDE (e.g., PyCharm, VSCode, or Jupyter Notebook).
 - Develop a simple program to display "Hello, World!" .
 - Demonstrate basic Python syntax and semantics.

Experiment 2: Basic Python Programming

- Objective: Learn basic programming constructs in Python.
- Tasks:
 - Develop programs using variables, data types, and operators.
 - Develop a program to implement basic input and output functions.
 - Develop programs using control structures (if statements, for loops, while loops).

Experiment 3: Functions and Modules

- Objective: Understand functions and module usage in Python.
- Tasks:
 - Define and call functions with different types of arguments and return values.
 - Develop a program to use built-in Python modules.
 - Develop a script that imports and utilizes at least two different standard library modules.

Experiment 4: Lists and Tuples

- Objective: Work with Python lists and tuples.
- Tasks:
 - Develop a program to create, modify, and iterate over lists and tuples.
 - Develop a program to perform list comprehensions to create new lists.
 - Demonstrate the immutability of tuples.

Experiment 5: Dictionaries and Sets

- Objective: Explore dictionaries and sets in Python.
- Tasks:
 - Develop a program to create and manipulate dictionaries.

- Develop a program to use dictionary comprehension.
- Develop a program to create and perform operations on sets.

- Experiment 6: Strings and File I/O

- Objective: Manipulate strings and perform file I/O operations.
- Tasks:
 - Demonstrate various string methods.
 - Develop programs to read from and write to text files.
 - Develop programs to work with different file formats, including CSV and JSON.

Experiment 7: Error Handling and Exceptions

- Objective: Implement error handling in Python programs.
- Tasks:
 - Develop programs using try, except, else, and finally blocks.
 - Develop programs to handle specific exceptions.
 - Develop programs to create and raise custom exceptions.

Experiment 8: Object-Oriented Programming (OOP)

- Objective: Understand and implement OOP concepts in Python.
- Tasks:
 - Develop programs to define classes and create objects.
 - Demonstrate inheritance and polymorphism.
 - Develop programs to use class and instance variables in programs.

Experiment 9: Libraries and Packages

- Objective: Utilize third-party libraries and create Python packages.
- Tasks:
 - Install and use libraries like NumPy and Pandas.
 - Create a simple Python package and distribute it.
 - Work with virtual environments to manage dependencies.

Experiment 10: Working with Data

- Objective: Perform data manipulation and visualization.
- Tasks:
 - Develop programs to use Pandas to load, manipulate, and analyze datasets.
 - Develop programs to create visualizations using matplotlib and seaborn.
 - Develop programs to conduct basic data analysis tasks and summarize findings.

Experiment 11: Web Scraping and APIs

- Objective: Extract data from the web and interact with APIs.
- Tasks:
 - Develop programs to access and parse data from RESTful APIs.
 - Develop programs to process and analyze JSON data from APIs.

Experiment 12: Databases

- **Objective:** Work with databases in Python.
- **Tasks:**
- Develop a program to connect to a database using SQLite and SQLAlchemy.
- Develop programs to perform CRUD operations on the database.
- Develop programs to manage and retrieve data using SQL queries.

References:

- https://www.udemy.com/course/python-the-complete-python-developercourse/?matchtype=e&msclkid=0584dfb54dc715f39c0bb9aaf74033be&utm_campaign=BGPpython_v.PROF_la.EN_cc.INDIA_ti.7380&utm_content=deal4584&utm_medium=udemysads&utm_source=bing&utm_term=._ag_1220458320107116_.ad_.kw_Python+language._de_c_.dm_.pl_.ti_kwd-76278984197882%3Aloc-90_.li_116074_.pd_.&couponCode=IND21PM
- https://www.w3schools.com/python/python_intro.asp
- <https://www.youtube.com/watch?v=eWRfhZUzrAc>
- https://onlinecourses.nptel.ac.in/noc20_cs83/preview
- <https://www.edx.org/learn/python>
- Virtual Labs - <https://python-iitk.vlabs.ac.in/>
- Virtual Labs - <https://virtual-labs.github.io/exp-arithmetic-operations-iitk/>
- Virtual Labs - <https://cse02-iiith.vlabs.ac.in/>
https://mlritm.ac.in/assets/cse/cse_lab_manuals/R20_cse_manuals/Python%20Lab%20Manual.pdf